

Numerical Experiments: Lévy Score-Corrected Compound-Poisson Sampling on Multimodal Benchmarks

Standalone experiment section with appendix

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Abstract

This document is a standalone rendering of the numerical-experiment section of the Lévy score-corrected compound-Poisson (LSC-CP) sampling paper, together with its appendix. Every number, table, and figure is generated from one audited production campaign (immutable run directories `20260717-prod-gpu7-{A2,B2,C3}`; commit `687c788`, working tree recorded dirty) that passed all fail-closed numerical gates: weak-stationarity certificates, timestep and quadrature refinement, nonfinite/clipping/basin-map audits, and reference diagnostics. Four systems are studied: the one-dimensional double well $V(x) = (x^2 - 1)^2$, a forty-mode planar Gaussian mixture, a depth-balanced three-well Müller-Brown landscape embedded in ten dimensions, and a coupled ϕ^4 chain in twenty-four dimensions. LSC-CP is the only method that reaches the finite-sample reference floor of the distributional metrics on all four systems while sustaining nonzero between-basin effective sample sizes; the three local baselines record exactly zero between-basin ESS on every system. All comparisons are equilibrium sampling diagnostics; no physical kinetics is claimed.

1 Overview and compared methods

All experiments target a Boltzmann law

$$\pi_\beta(x) = Z_\beta^{-1} e^{-\beta V(x)}, \quad x \in \mathbb{R}^d. \quad (1)$$

The sampler of interest augments overdamped Langevin dynamics with a finite compound-Poisson jump process of measure $\nu(dr) = \lambda\rho(dr)$ and adds the stationary Lévy-score drift $S_{\nu,\beta}$ so that the combined generator

$$A_{\nu,\beta}f = [-\nabla V + S_{\nu,\beta}] \cdot \nabla f + \beta^{-1} \Delta f + J_\nu f \quad (2)$$

is weakly stationary for π_β (Appendix A). The practical estimators evaluate the score integral by deterministic chord quadrature where affordable and by *paired* random-bank estimators in higher dimension: the realized jump displacements and the Lévy-score drift of one step are generated from the same random finite measure, so the drift-jump cancellation holds conditionally on every realized bank (Appendix A.1). Figures label the deployed estimator simply LSC-CP; run manifests record the exact variant. The deployed variants are: deterministic shell quadrature (double well), a closed-form score (MoG40), the paired single-atom estimator LSC-CP-RA (double well and MoG40, as an estimator-agreement check), and the paired multi-atom estimator LSC-CP-MA (Müller-Brown and ϕ^4).

Baselines are the unadjusted Langevin algorithm (ULA), Metropolis adjusted Langevin (MALA), the BAOAB underdamped integrator (BAOAB), an α -stable Lévy-flight analogue (FLA), and parallel

tempering (PT) with an automatically tuned ladder and fully counted replica cost. Raw compound Poisson (CP), i.e. the same jump bank *without* the score correction, is included as a stationarity-defect ablation: its invariant law is not π_β , so it is a control for the value of the correction, not a fair efficiency baseline. All methods share per-seed initial ensembles; CP and full-law LSC-CP additionally share the jump random stream pathwise.

Two claims are *not* made. First, no statement about physical reaction kinetics: the jump bank, score drift, taming, and algorithmic clock are designed for equilibrium sampling, and passage statistics compare algorithmic mixing only. Second, no claim that the finite-step chain preserves π_β exactly: the paired construction is exact at continuous-generator level, and the remaining finite- Δt error is measured by the refinement and audit gates of Section 3 rather than assumed away.

2 Benchmark systems

	Double well	MoG40	Depth-balanced MB	Coupled ϕ^4
dimension d	1	2	10	24
β	8	8	24	8
particles N	16,000	2,500	2,000	1,000
horizon T	100	100	200	100
Δt (refined)	0.005	0.010	0.005	0.002
steps	20,000	10,000	40,000	50,000
seeds	24	24	16	16
jump rate λ	1	1	1	1
score quadrature	$q_\theta=16, q_\rho=4$	closed form ($m_\phi=16$)	$q_\theta=8, q_\rho=4$	$q_\theta=32, q_\rho=4$
deployed estimator	exact + RA	closed form + RA	MA	MA
certificate residual	1.5×10^{-7}	3.0×10^{-8}	5.0×10^{-7}	2.0×10^{-7}

Table 1: Production configurations after fail-closed timestep and quadrature refinement, and the measured pre-production weak-stationarity certificate residual (gate: 10^{-6} ; Appendix E). Full model, bank, and ladder specifications are in Appendix B.

Double well ($d = 1$). $V(x) = (x^2 - 1)^2$ at $\beta = 8$: two equal wells at ± 1 separated by a dimensionless barrier $\beta\Delta V = 8$. The jump bank is a symmetric shell at ± 2 with radial half-width 0.2. This is the transparent two-state test of target recovery and interwell communication.

MoG40 ($d = 2$). π_β is an equal-weight mixture of forty unit-covariance Gaussians; $V(x) = -\beta^{-1} \log \sum_k \exp[-\|x - \mu_k\|^2/2]$. The jump law is deliberately generic: radius uniform on $[4, 15]$ (set from the nearest-neighbour distance histogram alone), direction uniform on the circle. Neither PT nor LSC-CP receives the mode locations. This is the many-basin stress test.

Depth-balanced Müller–Brown ($d = 10$). A Müller–Brown-type three-well landscape whose well depths are retuned to be equal, evaluated in a two-dimensional latent plane and embedded in ten dimensions by a fixed linear map with Gaussian auxiliary coordinates ($\sigma_{\text{aux}} = 0.4$), at $\beta = 24$. The equal depths separate multimodality from metastability: the channels retain dimensionless barriers $\beta b_{AB} \simeq 11.1$ and $\beta b_{BC} \simeq 4.0$ while the equilibrium masses stay comparable. The bank has four directed relay atoms $A \leftrightarrow B \leftrightarrow C$ (no direct A – C atom), so transport must use the relay through B . This tests curved, nonuniform basins and a designed jump graph.

Coupled ϕ^4 chain ($d = 24$). Twelve periodic two-component sites, $V(q) = \frac{\kappa}{2\delta} \sum_i \|q_{i+1} - q_i\|^2 + \delta \sum_i W(q_i)$ with $\kappa = 2.5$, $\delta = 1/12$, $\beta = 8$, and a tilted double-double-well site potential W (Appendix B). The target has four coherent phases; the bank has eight directed edge atoms along the phase square (diagonals excluded because their coherent chords cross the field-zero hilltop). The primary collective variable is the spatial mean $\bar{q} = \frac{1}{12} \sum_i q_i \in \mathbb{R}^2$. This is the many-body example: a collective transition in a coupled field.

The four systems form a hierarchy of equilibrium questions—two-state balance, many-mode occupancy, curved-basin relay transport, and a collective field transition—rather than four independent leaderboard tasks.

3 Protocol, references, and validation gates

Run protocol. For each system all seeds are batched into one ensemble per method with per-seed initial blocks; every method starts from the identical localized per-seed ensemble. The timestep is selected by a fail-closed halving rule: the largest dyadic Δt at which every π_β -targeting method’s terminal metrics agree with their $\Delta t/2$ values within declared floor-guarded tolerances (FLA and CP, whose invariant laws are not π_β , are recorded but do not gate). Score quadrature is selected by the analogous rule with an additional weak-stationarity certificate threshold. Exhausting either refinement grid aborts the run; it does not default to the finest attempted setting.

References and floors. The double well uses a dense-grid inverse-CDF reference; MoG40 an exact mixture draw; Müller–Brown a fine two-dimensional latent grid combined with exact Gaussian auxiliaries; ϕ^4 a Laplace-mixture proposal corrected by self-normalized importance weights (direct SNIS for basin masses, energy moments, CV means, and the free-energy surface; finite sampling-importance-resampling only where unweighted clouds are required). The ϕ^4 weight diagnostics for the production reference are: 2×10^5 proposals, weight ESS 41,182 (fraction 0.206), maximum normalized weight 3.6×10^{-3} (Appendix D). For every metric, a finite-sample *floor* is estimated by comparing two independent reference draws of the same size; “at the floor” below means within the floor band mean + 3 sd.

Gates. A run is accepted only if: the pre-production certificate residual is below 10^{-6} on a domain extending at least one maximum jump beyond the high-probability region; refinements pass; nonfinite counts are exactly zero; score-exponent and state-box clipping stay below declared tolerances; jump-boundary contact of the π_β -targeting jump methods stays below 10^{-2} per applied jump; and the basin-map-domain escape fraction of targeting methods stays below 10^{-3} . All thresholds, measured values, and the two domain-coverage design rules they enforce are reported in Appendix E. Every run writes its requested and resolved configurations, certificate, refinement tables, diagnostics, logs, and outputs into an immutable timestamped directory (Appendix F).

4 Equilibrium recovery

Four observations organize Table 2 and Figure 1.

(i) *Local samplers do not leave their initial basin.* On every system, ULA, MALA, and BAOAB remain essentially where they started for the entire horizon (basin TV 0.31–0.93, basin KL 1.1–4.2). At these dimensionless barriers this is expected and quantifies the problem the nonlocal methods must solve.

Method	W_2	TV_{basin}	$RMSE_F/k_B T$	$D_{\text{KL}}(p^* \ \tilde{p})$	FES outside
<i>Double well</i> (floors: W_2 0.061 ± 0.033 , TV 0.0032 ± 0.0017)					
ULA	1.26 (0.003)	0.471 (0.001)	1.77 (0.02)	1.09 (0.02)	0
MALA	1.26 (0.003)	0.473 (0.001)	1.79 (0.02)	1.12 (0.02)	0
BAOAB	1.29 (0.002)	0.488 (0.001)	2.23 (0.04)	1.54 (0.04)	0
FLA	0.096 (0.018)	0.0033 (0.003)	0.302 (0.008)	3.4×10^{-5}	0.003
PT	0.064 (0.032)	0.0033 (0.002)	0.050 (0.009)	3.2×10^{-5}	0
CP	0.176 (0.013)	0.0036 (0.003)	0.157 (0.006)	4.2×10^{-5}	0.029
LSC-CP	0.111 (0.017)	0.0036 (0.003)	0.168 (0.009)	4.4×10^{-5}	0.005
LSC-CP-RA	0.131 (0.017)	0.0031 (0.003)	0.180 (0.010)	3.2×10^{-5}	0.008
<i>MoG40</i> (floors: W_2 1.31 ± 0.28 , TV 0.051 ± 0.006)					
ULA	26.1 (0.02)	0.916 (0.003)	1.81 (0.01)	4.20 (0.02)	0
MALA	26.1 (0.03)	0.917 (0.003)	1.81 (0.01)	4.21 (0.01)	0
BAOAB	26.2 (0.03)	0.930 (0.005)	1.77 (0.01)	4.24 (0.02)	0
FLA	15.2 (0.21)	0.436 (0.007)	1.37 (0.06)	0.743 (0.05)	0.001
PT	6.54 (0.34)	0.141 (0.007)	0.350 (0.02)	0.055 (0.005)	0
CP	9.98 (0.19)	0.286 (0.009)	0.680 (0.04)	0.267 (0.013)	0.254
LSC-CP	2.83 (0.34)	0.064 (0.008)	0.190 (0.02)	0.012 (0.003)	0.001
LSC-CP-RA	2.65 (0.26)	0.103 (0.009)	0.280 (0.02)	0.036 (0.006)	0.005
<i>Depth-balanced MB, 10D</i> (floors: W_2 0.050 ± 0.020 , TV 0.011 ± 0.007)					
ULA	0.409 (0.003)	0.307 (0.002)	2.27 (0.23)	1.17 (0.14)	0
MALA	0.411 (0.003)	0.308 (0.001)	2.32 (0.18)	1.22 (0.16)	0
BAOAB	0.415 (0.003)	0.309 (0.001)	2.50 (0.17)	1.52 (0.21)	0
FLA	0.179 (0.006)	0.119 (0.008)	0.941 (0.04)	0.048 (0.007)	0.039
PT	0.056 (0.021)	0.013 (0.008)	0.132 (0.02)	5.7×10^{-4}	0
CP	0.353 (0.008)	0.135 (0.006)	0.267 (0.02)	0.047 (0.005)	0.172
LSC-CP-MA	0.047 (0.009)	0.014 (0.006)	0.136 (0.010)	5.8×10^{-4}	0.006
<i>Coupled ϕ^4, 24D</i> (floors: W_2 0.130 ± 0.035 , TV 0.024 ± 0.010)					
ULA	1.22 (0.004)	0.671 (0.003)	2.59 (0.09)	3.07 (0.31)	0
MALA	1.22 (0.004)	0.673 (0.002)	2.63 (0.05)	3.13 (0.17)	0
BAOAB	1.22 (0.006)	0.673 (0.002)	2.64 (0.05)	3.14 (0.22)	0
FLA	0.194 (0.032)	0.060 (0.013)	0.220 (0.03)	0.011 (0.004)	10^{-4}
PT	0.364 (0.028)	0.119 (0.015)	0.315 (0.04)	0.039 (0.008)	0
CP	0.410 (0.021)	0.077 (0.016)	0.482 (0.05)	0.017 (0.006)	0.209
LSC-CP-MA	0.100 (0.028)	0.022 (0.011)	0.151 (0.03)	0.0015 (0.001)	0.003

Table 2: Terminal metrics at $t = T$, mean (std) over seeds. W_2 : sliced (exact in 1D) Wasserstein-2 against the reference in metric space; TV_{basin} : total variation of basin occupancy; $RMSE_F$: additive-aligned free-energy-surface RMSE in $k_B T$ on the two-dimensional collective variables (one-dimensional for the double well); D_{KL} : target-to-empirical basin KL with Jeffreys pseudocount; FES outside: empirical mass outside the frozen reference FES grid. Metric definitions in Appendix C.

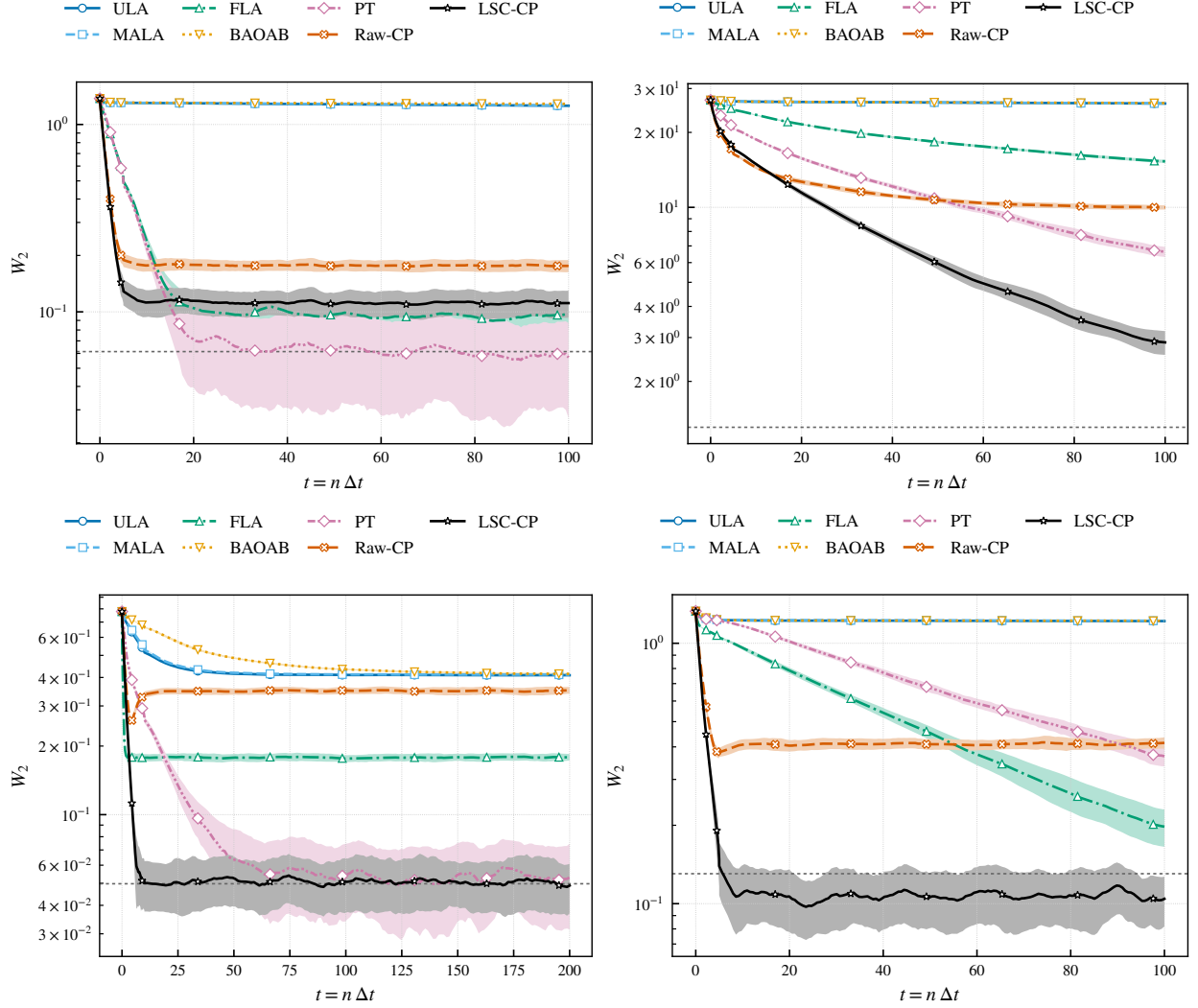


Figure 1: Ensemble relaxation: W_2 to the reference versus simulation time t from the shared localized initial ensembles (seed-band envelopes). Top: double well, MoG40; bottom: Müller–Brown 10D, coupled ϕ^4 .

(ii) *LSC-CP reaches the floor on all four systems.* The deployed LSC-CP estimator attains basin TV and KL at or within the finite-sample floor band on the double well, Müller–Brown, and ϕ^4 , and comes within 25% of the MoG40 TV floor while every alternative is at least $2.2\times$ further away. It is the only method with this property. Parallel tempering matches it on the double well and Müller–Brown but is $4.6\times$ worse in basin KL on MoG40 (0.055 vs 0.012) and $26\times$ worse on ϕ^4 (0.039 vs 0.0015), at full replica cost. The α -stable flight is competitive only on the double well; its invariant law is visibly biased elsewhere (MB free-energy RMSE $0.94 k_B T$).

(iii) *The score correction, not the jumps alone, closes the defect.* Raw CP shares the bank, jump times, and jump increments with LSC-CP pathwise, yet retains a large stationary defect: basin TV 0.286/0.135/0.077 on MoG40/MB/ ϕ^4 and, most tellingly, 17–25% of its terminal mass falls outside the reference free-energy grid on the three multivariate systems, versus $\leq 0.6\%$ for LSC-CP. Its boundary-contact rate—2.5%, 6.7%, and 0.4% of applied jumps on MoG40, MB, and ϕ^4 —is reported as part of this defect (Appendix E). We note for fairness that an earlier tight ϕ^4

integration box $[-2, 2]^{24}$ had *flattered* raw CP by confining its spread; with the audited box $[-5, 5]^{24}$ its W_2 moves from 0.28 to 0.41 while LSC-CP-MA is unchanged (0.099 vs 0.100), evidence that the corrected sampler is insensitive to the numerical box while the uncorrected one is not.

(iv) *Paired estimators agree with the deterministic score.* Where both are run (double well, MoG40), the paired single-atom estimator tracks the deterministic-quadrature score within a factor of 1.6 on every metric and with the same qualitative ordering (e.g. MoG40 basin KL 0.036 vs 0.012; double-well W_2 0.131 vs 0.111), supporting the use of the cheaper paired multi-atom estimator on the two systems where deterministic quadrature is unaffordable.

5 Stationary sampling efficiency

Terminal ensemble metrics answer “how close is the law at time T ”; they do not measure the precision of long-run time averages. For that, small independent chains (4 seeds $\times$ 8 chains, 1000 uniformly spaced draws over one production horizon) are initialized from the system reference, charged one full settling horizon, and analyzed for basin indicators, energy, and the collective variables. Every trace is labelled with its initialization provenance; none is claimed to be an exact stationary draw of the discrete kernel, so all values below are post-settling diagnostics of the numerical chains (Appendix C). Chains with zero observed transitions contribute zero ESS by construction; a trapped chain is reported as trapped, not as converged. Raw CP and FLA are excluded because their invariant laws are not π_β .

Method	worst-basin ESS (of 32,000)				energy ESS / signed energy bias			
	DW	MoG40	MB	ϕ^4	DW	MoG40	MB	ϕ^4
ULA	0	0	0	0	14,815 / +0.0003	707 / +0.001	22,223 / +0.007	17,212 / +0.133
MALA	0	0	0	0	14,869 / +0.0005	710 / -0.004	21,994 / -0.001	15,250 / -0.010
BAOAB	0	0	0	0	3,658 / -0.0002	534 / -0.001	5,622 / +0.004	2,734 / +0.028
PT	1,211	572	2,336	2,567	20,494 / -0.001	1,340 / +0.0006	25,202 / -0.0004	24,383 / -0.002
LSC-CP ^(*)	1,648	229	1,952	1,150	31,891 / +0.194	22,718 / +0.114	26,720 / +0.039	30,972 / +0.251
LSC-CP-RA	1,809	59	—	—	31,224 / +0.247	9,165 / +0.448	—	—

Table 3: Stationary-trace diagnostics aggregated over all seeds and chains. Worst-basin ESS is the minimum, over basins, of the summed per-chain effective sample size of that basin’s indicator (Geyer IAT; constant chains contribute zero). Energy bias is the signed difference between the trace mean of V and the reference target, in energy units; its magnitude should be read against the Monte Carlo standard error, which is below 0.006 in all cases. ^(*)LSC-CP row: deterministic/closed-form score on DW and MoG40, paired multi-atom on MB and ϕ^4 . All split- $\hat{R}$ values for the tabulated methods are below 1.11.

Three conclusions follow from Table 3.

(i) *Between-basin ESS separates the methods qualitatively.* The three local integrators record *exactly zero* between-basin ESS on every system: over one full production horizon at stationarity, no chain crossed a barrier even once. Their large energy ESS values illustrate why energy decorrelation alone must never be quoted as convergence evidence. LSC-CP sustains worst-basin ESS between 229 and 1,952—the same order as fully costed parallel tempering—using only local gradients plus the jump-bank score.

(ii) *Mixing and accuracy must be read together.* On MoG40 and ϕ^4 , PT shows the larger between-basin ESS (572 vs 229; 2,567 vs 1,150) yet its terminal occupancy accuracy is a factor 4.6–26 *worse* in basin KL (Table 2). Tempered swaps relabel basins quickly, but the resulting cold-chain law is

less accurate than the score-corrected chain at matched horizons. Neither column alone ranks the samplers; the product of accuracy and per-cost mixing does.

(iii) *The LSC chains carry a measurable, honest energy bias.* The stationary energy mean of the LSC chains exceeds the target by +0.04 to +0.25 energy units (many standard errors), while local and PT chains are unbiased at the 10^{-3} level. This is the expected signature of the finite-step chain’s invariant law: a jump deposits a particle on a high-energy chord point and the tamed drift relaxes it over a finite time, so an $O(\lambda\tau_{\text{relax}})$ fraction of stationary time is spent in elevated-energy transients. The distributional metrics of Table 2 include exactly these transients and still reach the reference floors, which bounds the practical impact; nevertheless every ESS or throughput number for LSC-CP must be displayed next to this bias, and is, throughout the released data.

Cost-normalized throughput. Per-second worst-basin throughput on the two cleanly timed systems (single tenant on one H200 GPU) is the same order for LSC-CP and PT: on MB, 33 vs 48 ESS/s (energy observable) with LSC-CP paying $Aq_\theta = 64$ score-quadrature energy differences per particle step; on ϕ^4 , 51 vs 62 ESS/s with $Aq_\theta = 256$. The double-well and MoG40 timing columns were collected while a second job shared the GPU and are reported in the manifests with that caveat; per-gradient, per-potential, and per-score-quadrature ESS, which are hardware independent, are stored per observable in the released stationarity tables (Appendix F).

6 Scope, negative controls, and outlook

Three boundaries of the present evidence are stated explicitly. First, all results are equilibrium-sampling diagnostics of algorithmic chains; passage or round-trip statistics, where recorded, are Kaplan–Meier restricted means of algorithmic mixing times and carry no molecular-kinetics meaning. Second, the finite-step LSC chains are not exactly π_β -invariant, and the measured consequences—the energy-transient bias of Section 5 and the audited clipping/boundary diagnostics of Appendix E—are part of the reported results rather than suppressed. Third, one planned ablation is not yet included: a matched-length *random-geometry* jump control (same atom count, length multiset, weights, rate, and cost; randomized directions) that would isolate the contribution of the designed bank geometry from the generic value of long jumps; raw CP controls for the score correction, not for geometry. A ϕ^4 box-sensitivity confirmation at $L = 6, 7$ and an independent (non-SNIS) ϕ^4 reference cross-check beyond the long PT run are planned for the same reason. These are completions of the control matrix, not changes to the sampler or claims.

A The Lévy score and the paired-bank estimators

For a finite jump measure ν , the jump generator is $(J_\nu f)(x) = \int [f(x+r) - f(x)] \nu(dr)$, and the Lévy score is

$$S_{\nu,\beta}(x) = - \int_{\mathbb{R}^d \setminus \{0\}} r \int_0^1 \frac{\pi_\beta(x - \theta r)}{\pi_\beta(x)} d\theta \nu(dr) = - \int r \int_0^1 e^{\beta[V(x) - V(x - \theta r)]} d\theta \nu(dr). \quad (3)$$

The chord fundamental theorem of calculus gives, for smooth compactly supported f ,

$$\int S_{\nu,\beta}(x) \cdot \nabla f(x) \pi_\beta(x) dx = - \int (J_\nu f)(y) \pi_\beta(y) dy, \quad (4)$$

hence $\int A_{\nu,\beta} f \pi_\beta dx = 0$ for the generator (2): an infinitesimal, continuous-time weak stationarity statement.

For a finite bank $\nu_{\mathcal{R}}(dr) = \lambda \sum_{a=1}^A w_a \delta_{R_a}(dr)$ the score is

$$S_{\mathcal{R},\beta}(x) = -\lambda \sum_{a=1}^A w_a R_a I_a(x), \quad I_a(x) = \int_0^1 e^{\beta[V(x) - V(x - \theta R_a)]} d\theta, \quad (5)$$

with I_a evaluated by Gauss–Legendre quadrature in θ (and, for the deterministic shell score, in the radial coordinate). The implementation accumulates the atom sum in log space with one *global* maximum exponent extracted after the signed-vector accumulation; capping that single global exponent preserves the relative atom weights, whereas per-atom capping would no longer compute the score of the realized measure. The cap activation fraction and maximum pre-cap exponent are recorded diagnostics (Appendix E).

A.1 Paired random banks

When the intended law is a mixture of conditional atom laws, $\nu(dr) = \lambda \sum_a w_a q_a(dr)$, the paired multi-atom estimator draws, at the start of each step and independently of the state,

$$R_{n,a} \sim q_a, \quad \widehat{\nu}_n(dr) = \lambda \sum_{a=1}^A w_a \delta_{R_{n,a}}(dr), \quad \mathbb{E}[\widehat{\nu}_n] = \nu, \quad (6)$$

evaluates the score (5) of the *realized* bank, and jumps with atomwise counts drawn conditionally on that same bank,

$$N_{n,a} \sim \text{Poisson}(\lambda w_a \Delta t), \quad \Delta J_n = \sum_{a=1}^A N_{n,a} R_{n,a}. \quad (7)$$

Conditional on every frozen bank, drift and jumps realize the same measure $\widehat{\nu}_n$, so (4) holds bank by bank at generator level; refreshing the state-independent bank averages back to ν . If instead the jumps were drawn from the marginal law while the score used an independently realized bank, the conditional defect $\int [J_{\widehat{\nu}} f - J_\nu f] \pi_\beta dx$ would generally be nonzero: unbiasedness of the score alone is not a substitute for conditional matching. The production sampler enforces the pairing structurally (the bank tensor is drawn once, used by both operators, then discarded) and its constructor rejects any score/jump configuration whose law object, rate, temperature, device, or precision disagree. The single-atom variant LSC-CP-RA is the $A = 1$ case with the atom index resampled each step.

The implemented step, with drift cap c set to twice the shell half-width and integration box $\mathcal{B}$, is

$$X_{n+1} = \text{clip}_{\mathcal{B}} \left[X_n + \Delta t \frac{b_n}{1 + \Delta t \|b_n\|/c} + \sqrt{2\Delta t/\beta} \xi_n + \sum_a N_{n,a} R_{n,a} \right], \quad b_n = -\nabla V(X_n) + S_{\mathcal{R}_n, \beta}(X_n). \quad (8)$$

Taming, clipping, quadrature, and the Euler step are outside the generator identity; their effects are measured (certificates, refinements, and the audit diagnostics of Appendix E), not assumed absent. Poisson multiplicities are applied exactly—never truncated—and multiple events on one atom within a step reuse its stored displacement, an $O(\Delta t^2)$ difference from full-law compound-Poisson sampling.

B System specifications

Double well. $V(x) = (x^2 - 1)^2$, $\beta = 8$; bank: atoms ± 2 , weights $1/2$, radial half-width 0.2 ; box $[-5.2, 5.2]$ (widened so that raw CP’s excursions do not pile mass at the edge of the density grid).

MoG40. Means μ_k drawn once (fixed seed) in $[-40, 40]^2$; $\pi_\beta = \frac{1}{40} \sum_k N(\mu_k, I_2)$; annulus law with radius uniform on $[4, 15]$, direction uniform; box $[-65, 65]^2$.

Depth-balanced Müller–Brown. Latent potential

$$V_{\text{MB3}}(z) = \sum_{k=1}^4 D_k \exp\{a_k(z_1 - x_k)^2 + b_k(z_1 - x_k)(z_2 - y_k) + c_k(z_2 - y_k)^2\}, \quad (9)$$

k	D_k	a_k	b_k	c_k	x_k	y_k
1	-1.6607	-1	0	-10	1	0
2	-1.0000	-1	0	-10	0	0.5
3	-1.0218	-6.5	11	-6.5	-0.5	1.5
4	0.1500	0.7	0.6	0.7	-1	1

(depths retuned so all three minima sit at $V \simeq -0.7957$), embedded as $x = zB^\top$ with eight auxiliary coordinates of scale $\sigma_{\text{aux}} = 0.4$; $\beta = 24$. Bank: four directed relay atoms $A \rightarrow B$, $B \rightarrow A$, $B \rightarrow C$, $C \rightarrow B$, weights $1/4$, radial half-width $0.1 \min_a \|r_a\|$; drift cap twice that half-width.

Coupled ϕ^4 . Site potential

$$W(u, v) = (u^2 - 1)^2 + (v^2 - 1)^2 - 0.0125 uv + 0.0075 u + 0.015 v, \quad (10)$$

$\kappa = 2.5$, $\delta = 1/12$, twelve periodic sites, $\beta = 8$. Bank: eight coherent edge atoms $\mathbf{1}_{12} \otimes (v_j - v_i)$ along the phase square, weights $1/8$, radial half-width $0.1 \min_a \|r_a\|$, drift cap twice that half-width. The integration box half-width $L = 5$ is *derived*: a simultaneous Laplace-component envelope with union-bound tail budget $\alpha = 10^{-8}$ gives $B_\beta = 1.910$, the hottest-PT envelope $B_{\beta_{\min}=1} = 3.569$, and the maximum componentwise one-jump reach $R_\infty = 2.200$; $L = \lceil \max\{B_\beta + R_\infty, B_{\beta_{\min}}\} \rceil = 5$. The basin map for $\bar{q}$ is constructed by gradient flow on $[-4, 4]^2$ —the *jump-reachable* order-parameter set (double-jump reach $|\bar{q}| \leq 3.3$), not merely the phase minima (Appendix E).

Parallel-tempering ladders (auto-tuned, fully costed). Double well: $\beta \in \{8, 1\}$ ($K = 2$, swap acceptance 0.41); MoG40: $\{8, 0.447, 0.025\}$ ($K = 3$, 0.26); MB: $\{24, 13.5, 7.6, 4.3\}$ ($K = 4$, 0.37); ϕ^4 : six geometric rungs from 8 to 1 ($K = 6$, 0.31).

C Metric definitions

Basin occupancy, TV, and KL. For basin partition $\{\Omega_k\}_{k=1}^K$ with target masses $p^\star$, $\hat{p}_k = \frac{1}{N} \sum_i \mathbf{1}\{X_i \in \Omega_k\}$, $\text{TV}_{\text{basin}} = \frac{1}{2} \sum_k |\hat{p}_k - p_k^\star|$, and

$$D_{\text{KL}}(p^\star \|\tilde{p}) = \sum_{k:p_k^\star > 0} p_k^\star \log \frac{p_k^\star}{\tilde{p}_k}, \quad \tilde{p}_k = \frac{\hat{p}_k + \alpha/N}{1 + K\alpha/N}, \quad \alpha = \frac{1}{2}. \quad (11)$$

The target-to-empirical orientation penalizes a sampler that misses a target basin; the Jeffreys-scale pseudocount (reported) keeps the statistic finite.

Free-energy-surface RMSE. On a fixed rectangular grid with cell probabilities p_i and volumes $|C_i|$, the reduced free energy is $A_i = -\log(p_i/|C_i|)$. With $\Delta_i = \hat{A}_i - A_i^\star$ on the supported cells $\mathcal{I} = \{i : p_i^\star \geq 5/N\}$ and reference weights $w_i = p_i^\star$,

$$\text{RMSE}_{F/k_B T} = \left[\frac{\sum_{\mathcal{I}} w_i (\Delta_i - c^\star)^2}{\sum_{\mathcal{I}} w_i} \right]^{1/2}, \quad c^\star = \frac{\sum_{\mathcal{I}} w_i \Delta_i}{\sum_{\mathcal{I}} w_i}, \quad (12)$$

i.e. free energies are compared after fitting the single additive constant they are defined up to. Grids use the two-dimensional active coordinates (one-dimensional for the double well), one-half pseudocount per cell, and the empirical mass outside the frozen grid is reported separately (Table 2) rather than renormalized away.

IAT, ESS, MCSE, $\hat{R}$. Per chain, the integrated autocorrelation time uses Geyer’s initial-positive/initial-monotone pairwise sums; a constant chain is assigned $\tau_{\text{int}} = \infty$ and zero ESS. Independent chains contribute $\text{ESS}_{\text{total}} = \sum_m n_m / \widehat{\tau_{\text{int}m}}$ (not draws over mean IAT). $\text{MCSE} = s/\sqrt{\text{ESS}}$, undefined for zero-variance traces. Split- $\hat{R}$ is rank normalized with average ranks for ties and Blom constant $(r-3/8)/(S+1/4)$, and the reported value is the maximum of the bulk and folded (median-absolute-deviation) statistics; globally constant traces give an undefined $\hat{R}$. Cost-normalized ESS divides by wall time, gradient evaluations, potential evaluations, and score-quadrature energy differences separately; observation functions run outside both the timer and the evaluation ledger.

Passage statistics. Where passage times are recorded they are right-censored at the common horizon, restricted to chains starting in the declared home core, and summarized by the Kaplan–Meier restricted mean; the model-dependent censored-exponential estimate, when saved, is labelled as such. Both are algorithmic-mixing diagnostics only.

D References and the ϕ^4 importance reference

Double well: dense-grid inverse CDF on the declared box (2×10^5 grid points). MoG40: exact mixture draws. Müller–Brown: fine latent grid (2400^2) times exact Gaussian auxiliaries. ϕ^4 : Laplace mixture at the four coherent minima (autograd Hessians) corrected by self-normalized importance weights. Production weight diagnostics: 2×10^5 proposals, weight ESS 41,182 (20.6%), entropy ESS consistent, maximum normalized weight 3.6×10^{-3} ; basin masses, energy moments, CV means, and the FES reference histogram use the weights directly (no resampling); finite SIR supplies unweighted clouds only where required (W_2 , MMD) and is labelled approximate. Reference floors in Table 2 are two-draw floors of the corresponding reference at matched sample size.

E Numerical gates, audit diagnostics, and measured values

Diagnostic (per run; gate)	DW	MoG40	MB	ϕ^4
nonfinite states (gate = 0)	0	0	0	0
score-cap activation (gate $\leq 10^{-2}$)	0	0	0	0
state-box clip fraction (gate $\leq 10^{-2}$)	1.7×10^{-6}	3.2×10^{-4}	2.5×10^{-3}	9×10^{-6}
jump-boundary contact, LSC (gate $\leq 10^{-2}$)	5.3×10^{-4}	1.1×10^{-4}	2.2×10^{-3}	1.6×10^{-4}
jump-boundary contact, raw CP (recorded)	3.0×10^{-4}	2.5×10^{-2}	6.7×10^{-2}	4.1×10^{-3}
basin-map escape, targeting (gate $\leq 10^{-3}$)	0	0	0	1.0×10^{-3}
certificate residual (gate $\leq 10^{-6}$)	1.5×10^{-7}	3.0×10^{-8}	5.0×10^{-7}	2.0×10^{-7}

Table 4: Fail-closed gate thresholds and the values measured by the accepted production runs. Jump-boundary contact is per applied jump; state-box clipping per proposal (the largest value belongs to the heavy-tailed FLA baseline). The ϕ^4 basin-map entry corresponds to a single particle at a single checkpoint ($1/N$ with $N = 1000$).

Two domain-coverage rules were established by this campaign’s failed first attempts, and are now part of the audit design.

Rule 1: a hard-zero jump-boundary gate is unsatisfiable. CP-family jumps fire independently of the state, so rare multi-jump excursions contact *any* finite integration box; measured LSC contact rates are 10^{-4} – 2×10^{-3} per applied jump. The gate therefore bounds π_β -targeting methods at 10^{-2} (five to fifty times the observed values) and *records* raw CP, whose larger contact rate is part of its documented stationarity defect—consistent with the refinement and basin gates, which likewise never gate on non-targeting baselines. Box truncation flatters raw CP (Section 4), so recording it is conservative for the comparison.

Rule 2: every domain-restricted diagnostic must cover the jump-reachable set. The ϕ^4 basin map was originally built on the phase-minima square $[-2, 2]^2$; a single phase-to-phase atom moves $\bar{q}$ by ≈ 2.12 , so legitimate LSC-CP transport parks $|\bar{q}|$ up to 3.3, which the map lookup would silently clamp into boundary cells and mislabel. The map is now built on $[-4, 4]^2$ at unchanged cell size, with the metric bounds derived from the map object itself so the gate and the lookup can never measure different domains.

F Provenance and runtime

Accepted runs: 20260717-prod-gpu7-A2 (MoG40), 20260717-prod-gpu7-B2 (double well, Müller–Brown), and 20260717-prod-gpu7-C3 (ϕ^4), each an immutable directory containing the requested YAML, resolved preflight and post-refinement JSON configurations with cache SHA-256 values, the certificate, refinement tables, metric time series, per-method summaries, stationarity traces (NPZ) and tables (CSV), figures, logs, and a status manifest; git commit 687c788 with the working tree recorded dirty (the audited revision under review). Hardware: one NVIDIA H200 (float64 throughout), PyTorch 2.13.0+cu130, Python 3.12.13. End-to-end notebook wall times: double well 142 min, MoG40 134 min, Müller–Brown 91 min, ϕ^4 80 min, including references, certificates, refinements, production dynamics, stationary traces, and rendering. The double-well and MoG40 runs shared the GPU with each other for part of their execution; their wall-clock-derived columns carry that caveat in the manifests (the co-tenant process list at start is recorded), while the Müller–Brown and ϕ^4 runs were single tenant. Hardware-independent cost counters (gradient, potential, and score-quadrature evaluations) are recorded per method and per observable in all cases. Two

preceding campaign attempts failed post-production audit gates (the jump-boundary and basin-map-domain issues of Appendix E); their run directories and failure manifests are retained.